

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	1 NOVEMBER 2025
Team ID	NM2025TMID07302
Project Name	Calculating Family Expenses using Service Now
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Architecture Overview:

The “Calculating Family Expenses Using ServiceNow” system follows a simple cloud-based architecture built entirely on the ServiceNow platform.

- User Interface Layer: Family members use the ServiceNow web dashboard to enter and view their expenses. Each user logs in securely and can access only their permitted data.
- Application Logic Layer:The business logic is implemented using Flow Designer and server-side scripts. This layer handles expense calculations, validations, and automation of workflows.
- Database Layer:All expense details, user profiles, and budget limits are stored in ServiceNow’s CMDB and custom tables. The data is managed securely in the ServiceNow Cloud Database.
- Notification Layer:When expenses exceed set limits, automated notifications are triggered and sent to users via email or in-app messages.
- Integration Layer (Optional):The system can connect with third-party financial tools or APIs for analytics or currency conversion if needed.
- Infrastructure Layer:The entire application is hosted on the ServiceNow SaaS Cloud, ensuring scalability, high availability, and data security.

Architecture Overview

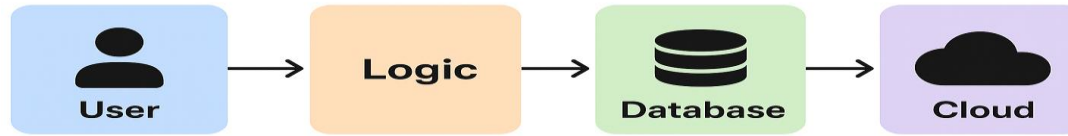


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Admin interacts via Web dashboard in ServiceNow.	ServiceNow Web UI
2.	Application Logic-1	Checks if user is linked to an incident before allowing deletion	ServiceNow Flow Designer, Script
3.	Application Logic-2	Validates user status from incident table	GlideRecord in Server Script
4.	Application Logic-3	Sends notification if deletion is blocked.	ServiceNow Notifications
5.	Database	Stores user, incident, and assignment data	ServiceNow CMDB, Incident Tables
6.	Cloud Database	Managed by ServiceNow cloud backend.	ServiceNow Cloud Database
7.	File Storage	Minimal use; logs stored internally	ServiceNow system logs
8.	External API-1	(Optional) HRMS integration to verify user status	REST API in ServiceNow
9.	External API-2	Not applicable	-
10.	Machine Learning Model	Not applicable for current use case	-
11.	Infrastructure (Server / Cloud)	Hosted and managed on ServiceNow SaaS platform	ServiceNow Cloud (SaaS)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Not applicable (ServiceNow is proprietary)	-
2.	Security Implementations	Role-based access control, ACLs, secure scripts	ACLs, Scoped Applications
3.	Scalable Architecture	SaaS-based, horizontally scalable via ServiceNow architecture	ServiceNow Cloud Architecture
4.	Availability	Highly available with ServiceNow cloud hosting	Load-balanced ServiceNow Instances
5.	Performance	Optimized via asynchronous flows and indexed tables	GlideRecord, Background Scripts